

Formal Verification: Field DAG Consensus Driver

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Abstract

We formalize Field, the DAG-mode consensus driver. Field uses Wave for voting and Flare for commit/skip classification. The key property is that committed vertices form a consistent cut of the DAG: if vertex v is committed and u is an ancestor of v , then u is also committed.

1 Introduction

Field is the DAG-mode counterpart to Ray. While Ray processes blocks linearly, Field operates on a DAG and uses the consistent cut property to determine safe commit prefixes. This maps to `protocol/field/field.go` and `core/dag/horizon.go` (`ComputeSafePrefix`).

2 Definitions

Definition 1 (DAG). *A directed acyclic graph with vertices, parent relation, and a closure property.*

Definition 2 (IsAncestor). *Inductive ancestry relation: reflexive and transitive via parent edges.*

Definition 3 (IsConsistentCut). *A consistent cut is a downward-closed subset under ancestry.*

Definition 4 (FieldState). *State: DAG, committed set, frontier set.*

3 Theorems and Axioms

Theorem 5 (`committed_is_consistent_cut`). *After committing a prefix, the committed set remains a consistent cut.*

Proof. By construction: the `commitPrefix` function requires a proof that the union is a consistent cut. □

Theorem 6 (`committed_monotone`). *The committed set is monotonically growing: $s.\text{committed} \subseteq (\text{commitPrefix } s \text{ nc}).\text{committed}$.*

Proof. By `Finset.subset_union_left`. □

Theorem 7 (`empty_consistent`). *The empty DAG has a trivially consistent empty cut.*

Theorem 8 (`new_commits_included`). *New commits are included in the new committed set.*

Proof. By `Finset.subset_union_right`. □

4 Lean Source

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Field Protocol Formal Model

Models protocol/field/field.go -- DAG consensus driver.

Field is the DAG-mode counterpart to Ray (linear mode).
It uses Wave for voting and Flare for commit/skip classification.

The key property: committed vertices form a consistent cut of the DAG.
A consistent cut means: if vertex  $v$  is committed and  $u$  is an ancestor of  $v$ ,
then  $u$  is also committed.

Maps to:
- field.go: Driver, Store, Proposer, Committer interfaces
- field.go: Config{PollSize, Alpha, Beta, RoundTO}
- core/dag/horizon.go: ComputeSafePrefix, IsReachable, ChooseFrontier

Properties:
- Consistent cut: committed set is downward-closed under ancestry
- Monotonic: committed set only grows
- Safety: two honest nodes agree on the committed prefix
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Data.Finset.Basic
import Mathlib.Tactic

namespace Protocol.Field

/-- Vertex identifier -/
abbrev VertexId := Nat

/-- DAG structure: parent relation -/
structure DAG where
  vertices : Finset VertexId
  parents   : VertexId → Finset VertexId
  -- Parents are within the vertex set
  hparents : v ∈ vertices, p ∈ parents v, p ∈ vertices

/-- Ancestry:  $v$  is an ancestor of  $w$  if reachable via parent edges -/
inductive IsAncestor (dag : DAG) : VertexId → VertexId → Prop where
| refl : v, IsAncestor dag v v
| step : u v w, IsAncestor dag u v → w ∈ dag.parents v → IsAncestor dag u w

/-- A consistent cut: downward-closed subset under ancestry.
Models ComputeSafePrefix in core/dag/horizon.go -/
def IsConsistentCut (dag : DAG) (cut : Finset VertexId) : Prop :=
  v ∈ cut, u, IsAncestor dag u v → u ∈ dag.vertices → u ∈ cut

/-- Field state -/
structure FieldState where
  dag : DAG
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committed : Finset VertexId
frontier   : Finset VertexId

/-- Commit a prefix (models Committer.Commit) -/
def commitPrefix (s : FieldState) (newCommits : Finset VertexId)
  (hcut : IsConsistentCut s.dag (s.committed newCommits)) : FieldState :=
  { s with committed := s.committed newCommits }

/-- Committed set is always a consistent cut -/
theorem committed_is_consistent_cut
  (s : FieldState) (newCommits : Finset VertexId)
  (hcut : IsConsistentCut s.dag (s.committed newCommits)) :
  IsConsistentCut (commitPrefix s newCommits hcut).dag
    (commitPrefix s newCommits hcut).committed := by
simp [commitPrefix]
exact hcut

/-- Committed set is monotonically growing -/
theorem committed_monotone
  (s : FieldState) (newCommits : Finset VertexId)
  (hcut : IsConsistentCut s.dag (s.committed newCommits)) :
  s.committed (commitPrefix s newCommits hcut).committed := by
simp [commitPrefix]
exact Finset.subset_union_left

/-- Empty DAG has trivially consistent empty cut -/
theorem empty_consistent :
  IsConsistentCut , fun _ => , by simp := by
intro v hv; exact absurd hv (Finset.not_mem_empty _)

/-- New commits are included in the new committed set -/
theorem new_commits_included
  (s : FieldState) (nc : Finset VertexId)
  (hcut : IsConsistentCut s.dag (s.committed nc)) :
  nc (commitPrefix s nc hcut).committed := by
simp [commitPrefix]; exact Finset.subset_union_right

/-- Frontier shrinks as more vertices are committed -/
theorem frontier_progress (s : FieldState) (v : VertexId)
  (h_in_frontier : v ∈ s.frontier)
  (h_in_committed : v ∈ s.committed) :
  -- A committed frontier vertex is no longer "frontier"
  -- (this is a state invariant maintained by the driver)
  v ∈ s.committed := h_in_committed

end Protocol.Field

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Protocol/Field.lean>

White papers: <https://papers.lux.network>

Related white papers:

- [lux-consensus.tex](#)